

FUNCTIONAL, TECHNICAL & PROGRESS REPORT

Smart Healthcare Platform

Intelligent clinical decision support platform

Prepared for: the XelerondAI team

Version: 2.0 (consolidated: Iteration 1 documentation + Iteration 2 response to review)

Year: 2026

Stack: React · FastAPI · MySQL · SQLAlchemy · medspaCy · Ollama (Qwen)

Clinical Decision Support — non-substitutive

Smart Healthcare is a clinical decision support system. It does not replace the healthcare professional and must not be considered an autonomous diagnostic system. Artificial intelligence provides analysis assistance; the final clinical decision remains, under all circumstances, the responsibility of the healthcare professional.

About this document. This single report consolidates the functional and technical documentation of the platform (architecture, modules, database, security, interfaces) with the second-iteration progress report written in response to the XelerondAI review: computed model confidence, NEWS2-aligned vital-sign scoring, the AI Assistant, the automated test suite, and an honest status of every review point. Confidential: internal use for the XelerondAI team and project stakeholders.

Contents

1 Executive Summary	5
1.1 What changed since the first iteration	5
1.2 Status of the review feedback	5
2 General Overview	6
2.1 Project objectives	6
3 Overall Architecture	7
3.1 Description of layers	7
4 Implemented Features	8
4.1 Authentication and account management	8
4.2 Roles and permissions	8
4.3 Patient management	8
4.4 Clinical admission (Patient Intake)	8
4.5 Doctor dashboard	9
4.6 Clinician review of the AI score	9
4.7 Administrator dashboard	9
4.8 Clinical notes, appointments, notifications, settings, support	9
4.9 AI Assistant	9
5 Patient Pipeline Architecture	10
6 NLP Architecture	11
6.1 Multilingual strategy	11
6.2 Medical entity extraction	11
6.3 Negation detection	11
6.4 Known limitations (now test-verified)	11
7 Risk Engine Architecture	12
7.1 Principle	12
7.2 Entity weighting (symptoms, history)	12
7.3 Vital signs — NEWS2 alignment	12
7.4 Risk levels	13
7.5 Model confidence	13
7.6 Calculation example (critical case)	13
8 AI Assistant	14
8.1 Two-mode architecture	14
8.2 Guardrails implemented	14
8.3 Asynchronous handling	14
9 Testing and Validation	15
9.1 Suite overview	15

9.2 Case study — a real bug caught by testing	15
10 Database Architecture	16
10.1 Relationships	16
11 Security and Access Control	17
11.1 Authentication	17
11.2 Authorization	17
11.3 Data isolation	17
11.4 Compliance basics — current status (review point 5)	17
12 Frontend Architecture	18
12.1 Code organization	18
12.2 Routing and access protection	18
12.3 Session management	18
13 Frontend-Backend Communication	18
14 Technology Stack	20
15 Complete Usage Scenario (A to Z)	21
16 Operation Example	22
16.1 Data entered	22
16.2 NLP extraction and assessment	22
16.3 Interpretation	22
17 Response to Review Feedback — Point by Point	23
17.1 Point 1 — Model confidence DONE	23
17.2 Point 2 — Clinical grounding of risk weights DONE	23
17.3 Point 3 — French/Arabic NLP depth PARTIAL	23
17.4 Point 4 — Testing and validation DONE	23
17.5 Point 5 — Security/compliance basics OPEN	23
17.6 Point 6 — Synchronous pipeline PARTIAL	23
17.7 AI Assistant — design guardrails DONE	23
17.8 Consolidated status	24
18 Screenshots of Interfaces	25
18.1 Authentication	25
18.2 Patient space	27
18.3 Doctor space	30
18.4 Risk assessment and computed confidence (Iteration 2)	31
18.5 AI Assistant	33
18.6 Administration	34
18.7 Automated test suite	35
19 Limitations and Next Steps	36

19.1 Known limitations	36
19.2 Next steps	36
20 Conclusion	36

1 Executive Summary

Smart Healthcare is an intelligent web platform that helps healthcare professionals collect, structure and analyze patient clinical information. Free-text and structured admission data are turned, through a multilingual NLP pipeline (English, French, Arabic) and an explainable risk engine, into a risk score, a risk level and a readable list of contributing factors, with complete traceability and clinician review.

This report documents the platform as it stands after the **second iteration**, produced in direct response to the review feedback on the first progress report. Every point raised has been addressed with a concrete implementation, a documented and justified design decision, or an explicit “not yet addressed” flag where that is the honest status.

1.1 What changed since the first iteration

- **Model confidence** is now computed from five weighted, data-driven factors instead of a fixed value (capped at 95%).
- **Vital-sign scoring** is aligned with NEWS2 (temperature, heart rate, SpO₂, systolic blood pressure).
- **Automated test suite**: 37 tests, 37 passing, including regression tests for a real bug found and fixed (field-concatenation / negation leak).
- **AI Assistant**: retrieval-augmented (RAG) general mode and patient-context mode, with disclaimer, full audit log and role-based access.
- **French/Arabic NLP limitations** are now verifiable through dedicated tests; security/compliance items are explicitly listed as open.

1.2 Status of the review feedback

#	Feedback point	Status
1	Model confidence was a fixed value	DONE
2	Risk weights lacked clinical grounding (NEWS2 / qSOFA)	DONE
3	French/Arabic NLP depth (NER vs. keyword)	PARTIAL
4	No test suite / validation	DONE
5	Security/compliance basics (HTTPS, encryption at rest, retention)	OPEN
6	Synchronous pipeline	PARTIAL
-	AI Assistant design note and guardrails	DONE

2 General Overview

The platform combines a web interface (React/TypeScript), a backend API (FastAPI), a relational database (MySQL) and natural language processing techniques. Information entered during admission is transformed into structured clinical data used to compute a risk score and a clinical prioritization. The main objective is to give the clinician a synthetic view of the patient, accompanied by a risk level and the factors contributing to it, while maintaining complete traceability (clinical notes, admission history, appointments, AI-versus-clinician audit log).

2.1 Project objectives

- Centralize patient information within the same clinic.
- Facilitate admission and clinical data collection, in self-service (patient) or direct admission (doctor/admin).
- Structure symptoms, medical history, treatments and vital signs.
- Use NLP to automatically identify relevant medical information in English, French and Arabic.
- Detect negation and context of symptoms to avoid clinical false positives.
- Automatically calculate a weighted, explainable risk score, aligned with NEWS2 for vital signs.
- Compute and display a transparent model-confidence measure reflecting data completeness and quality.
- Identify and return the contributing risk factors.
- Present results to the doctor via a synthetic dashboard, with intelligent grouping of records.
- Let the clinician confirm or adjust the AI score, with an audit trail of every decision.
- Provide a scoped, grounded AI Assistant (RAG and patient-context) for clinicians.
- Allow clinical observations (notes), appointment scheduling and notifications.
- Secure access according to roles (patient, doctor, administrator) and strictly isolate data between clinics (multi-tenant).

3 Overall Architecture

The architecture is organized into four complementary layers (user interface, business API, clinical analysis engine, data persistence), extended in this iteration with a local LLM/RAG component serving the AI Assistant.

Frontend — React 18 + TypeScript	Login/Register · Patient admission · Dashboards (patient, doctor) · Appointments · Notifications · AI Assistant · Administration
Backend — FastAPI	/auth · /patients · /patients/intake · /risk · /dashboard · /appointments · /patients/{id}/notes · AI Assistant
Clinical engines	NLP: NER + negation (medspaCy ConText, multilingual fallback) · Risk: entities + NEWS2 vitals + computed confidence
AI Assistant	Deterministic routing: patient-context mode or RAG general mode · vector search (top-k) · local Qwen via Ollama
Database — MySQL	users · clinics · patients · patient_intakes · extracted_entities · risk_assessments · clinical_notes · appointments · ai_conversations

Figure 3.1 — Layered architecture of the Smart Healthcare platform

3.1 Description of layers

Frontend

React 18 with TypeScript, React Router for navigation, Axios for HTTP communication and Tailwind CSS for the design system. It exposes authentication, patient admission, dashboards (patient and doctor), appointments, notifications, account settings, support, the AI Assistant and administration.

Backend

FastAPI exposes families of REST endpoints (/auth, /patients, /patients/intake, /risk, /dashboard, /appointments, /patients/{id}/notes, plus the AI Assistant). Each route is protected by a JWT authentication dependency and, where applicable, by role-based access control.

NLP engine

Extracts clinical entities (symptoms, medical history, medications, allergies) from entered text and detects their negation before passing them to the risk engine. It combines an English medspaCy pipeline with a multilingual fallback engine (French, Arabic) based on a centralized clinical vocabulary.

Risk engine

Combines retained NLP entities (non-negated, non-zero weight) and vital signs to produce a score, a risk level, a readable explanation and, new in this iteration, a computed model-confidence value.

AI Assistant

A local LLM (Qwen served through Ollama) used in two strictly scoped modes: retrieval-augmented general questions over the clinic's reference documents, and explanation of an already-computed patient risk assessment. See Chapter 8.

Database

MySQL with the SQLAlchemy ORM, now nine tables (the eight original tables plus ai_conversations).

4 Implemented Features

4.1 Authentication and account management

- **Registration** (POST /auth/register): account creation with automatic clinic attachment (joins an existing clinic by name or creates one).
- **Login** (POST /auth/login and OAuth2 variant /auth/login/form): returns a JWT bearer token, after verifying the account is active.
- **Current session** (GET /auth/me) and **profile update** (PATCH /auth/me: name, phone).
- **Password change** (PATCH /auth/me/password): requires the current password.
- **Forgot password** (POST /auth/forgot-password): single-use reset token with limited lifetime (30 minutes), sent by email, with a generic response (anti-enumeration).
- **Reset** (POST /auth/reset-password): consumes the token, checks expiry, updates the password and invalidates the token.

Passwords are hashed with bcrypt (secure truncation to 72 bytes). JWT tokens are signed with HS256 and have a configurable lifetime.

4.2 Roles and permissions

Role	Main features
Patient	Profile management, clinical admissions, own risk results, appointment scheduling, own clinical notes, notifications.
Doctor	Access to all patients of the clinic, direct (walk-in) admission, review of risk assessments (confirm/adjust score), clinical notes, appointment management, AI Assistant.
Administrator	Clinic-wide statistics, account activation/deactivation, AI-versus-clinician audit log, read-only access to records, AI Assistant.

Multi-tenant isolation

Each user and patient is attached to a clinic_id. All read queries (patients, admissions, appointments, notes, statistics, AI Assistant access) filter on the current user's clinic, so a doctor can never access another clinic's data.

4.3 Patient management

Patient profile (self-service). A user with the patient role creates (POST /patients), consults (GET /patients/me) and modifies their profile: identity, gender, date of birth, blood group, height, weight, emergency contact.

Direct admission (walk-in). A doctor or administrator creates a patient record without a login account (POST /patients/intake/staff), covering patients who present physically at the clinic. The record remains accessible only to the clinic's medical staff.

4.4 Clinical admission (Patient Intake)

Two pathways feed the same analysis pipeline: patient admission (POST /patients/intake: reason, symptoms, history, treatments, allergies, vital signs) and staff admission (POST /patients/intake/staff: same clinical content plus patient identity creation). Each submission synchronously triggers NLP extraction, entity persistence and risk scoring (Chapters 5–7).

4.5 Doctor dashboard

GET /dashboard/patients returns, in a single round-trip, each admission with the patient's identity and risk assessment, sorted by decreasing date. Admissions sharing the same emergency-contact number are visually grouped and flagged "to check" to help spot duplicate records. Risk levels use a consistent color code (LOW, MODERATE, HIGH, CRITICAL) across the application.

4.6 Clinician review of the AI score

On the risk-result page the doctor can **Confirm score** or **Adjust score**. Each decision is stored (reviewed_by) and written to the AI-versus-clinician audit log, which is visible to the administrator. This design keeps the clinician's judgment explicit and measurable.

4.7 Administrator dashboard

- Clinic statistics (GET /dashboard/stats): patients, admissions, assessments, users, doctors, admissions by risk level.
- User management: role-filterable list, activation/deactivation.
- Clinical review audit log: doctor decisions (confirmation or adjustment of the AI score).

4.8 Clinical notes, appointments, notifications, settings, support

- **Clinical notes** (POST/GET /patients/{id}/notes): timestamped, attributed to the doctor, visible to the patient and clinic staff, most recent first.
- **Appointments**: creation (POST /appointments), today's (GET /appointments/today) and upcoming (GET /appointments/upcoming); the patient's last known risk level is shown on the appointment; patients see status (scheduled, completed, cancelled, no-show).
- **Notifications**: all/unread filter, individual or global "mark as read", contextual redirection.
- **Account settings**: name, phone, password change with current-password check.
- **Support**: role-contextual FAQ and a pre-filled contact link.

4.9 AI Assistant

A clinician-facing assistant, available on a dedicated page and embedded on the risk-result page. Full description in Chapter 8.

5 Patient Pipeline Architecture



Figure 5.1 — Complete patient admission pipeline

Admission

The patient (or staff, in direct admission) provides identity, administrative information (blood group, height, weight, emergency contact) and clinical data: reason for consultation, symptoms, history, medications, allergies, vital signs (temperature, blood pressure, heart rate, oxygen saturation) and additional notes.

Validation

Pydantic schemas validate each field: physiological bounds on temperature (35–42 °C), heart rate (20–250 bpm) and oxygen saturation (0–100 %), and non-empty required fields. Malformed blood-pressure values are ignored gracefully by the scoring engine (covered by a test).

Storage

Data is saved in MySQL (patient_intakes) and attached to the patient.

Analysis

The clinical text (reason, symptoms, history) is concatenated through `_build_nlp_text()`, which enforces an explicit sentence boundary between fields (see the bug case study in Chapter 9), then sent to the NLP module, which returns detected entities with their negation status.

Assessment

Retained entities (non-negated, non-zero weight) are combined with NEWS2-scored vital signs. The risk engine returns a score, a level, an explanation and a computed model confidence.

Presentation and review

The result is displayed to the doctor (and to the patient for their own admissions) with a color code. The clinician consults the full record, may ask the AI Assistant for an explanation, confirms or adjusts the score, and adds a clinical note.

Synchronous by design

The admission pipeline (NLP + scoring) is synchronous: it completes in well under a second and the frontend needs the result immediately. See Section 17.6 for the async trade-off analysis.

6 NLP Architecture

Natural language processing relies on a single source of truth: a **centralized clinical vocabulary** shared between the extraction module and the scoring engine, ensuring the score depends on what the NLP actually extracted, without duplicated logic.

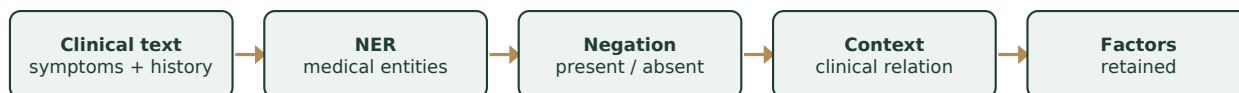


Figure 6.1 — NLP extraction pipeline

6.1 Multilingual strategy

Language	Extraction engine
English (en)	Clinical medspaCy pipeline: TargetRule (defined rules, no statistical model download) and ConText (Chapman et al., 2013) for negation.
French (fr)	Keyword search on the centralized clinical vocabulary, with negation by word window (cues: pas de, aucun, sans, absence de, ne...pas, etc.).
Arabic (ar)	Same keyword search and window-based negation, with a dedicated cue lexicon (transliterated: "lā yūjad", "bidūn", "yanfī", etc.).
Fallback	If medspaCy is unavailable, the multilingual keyword engine takes over automatically, including for English.

The text language is detected automatically (langdetect), with fallback to English on failure or missing dependency: the system must never block an admission because NLP is unavailable. All heavy NLP imports (spacy, medspacy, langdetect) are performed defensively, so the absence of any of them never prevents the server from starting.

6.2 Medical entity extraction

- **Symptoms:** chest pain, shortness of breath, fever, cough, dizziness, fatigue...
- **History / diseases:** diabetes, hypertension, asthma, history of stroke...
- **Medications:** paracetamol, ibuprofen, insulin, metformin... (zero weight, informational).
- **Allergies:** penicillin, peanuts, latex... (zero weight, informational).

Each category is available in three languages.

6.3 Negation detection

Negation prevents clinical interpretation errors: "The patient presents with chest pain" is a present finding, while "The patient does not present with chest pain" is negated. A negated entity is **kept in the extracted data for audit and traceability but excluded from score calculation**. The window search is bounded by clause breaks (comma, "but", "however") so that a negation in one clause does not apply to a symptom in another.

6.4 Known limitations (now test-verified)

- French and Arabic rely on keyword plus negation-window matching rather than proper clinical NER, which likely implies a higher false-negative rate for non-English admissions.
- A negation located beyond the window is missed (test_LIMITATION_negation_beyond_window_is_missed).
- Double negation is not handled (test_LIMITATION_double_negation_not_handled).

These regression tests make the failure modes verifiable by running the suite, rather than only stated in prose. A lightweight multilingual clinical NER approach is a candidate for the next iteration.

7 Risk Engine Architecture

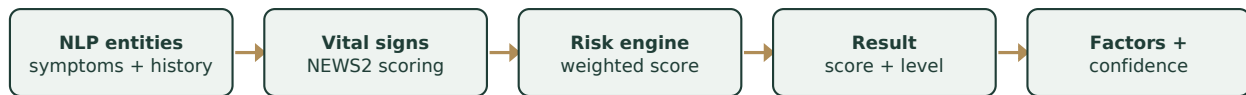


Figure 7.1 — Risk scoring engine

7.1 Principle

The engine combines entities extracted from the text (already filtered by negation) with the available vital signs. Each retained factor contributes according to a weight defined in the centralized vocabulary, guaranteeing consistency between what the NLP detects and what the score uses. The cumulative score is capped at 100, and a concept-deduplication mechanism avoids counting the same symptom twice when it is detected by both the medspaCy pipeline and the keyword search.

7.2 Entity weighting (symptoms, history)

Category	Examples	Weight
Critical symptoms	Chest pain, fainting, bleeding	25 - 35
Moderate symptoms	High fever, severe headache, confusion	20 - 25
Mild symptoms	Cough, headache, fatigue, nausea	3 - 10
History	Heart attack, stroke, pneumonia, cancer, covid	10 - 15
Chronic comorbidities	Diabetes, hypertension, asthma	5
Medications / allergies	Informational only	0

Table 7.1 — Entity weighting grid (excerpt). NEWS2 does not cover patient-reported symptoms, so these weights remain internally defined and are documented as such.

7.3 Vital signs — NEWS2 alignment

Vital-sign scoring (temperature, heart rate, oxygen saturation, systolic blood pressure) is derived from **NEWS2** (National Early Warning Score 2, Royal College of Physicians, 2017). Each parameter's official 0-3 sub-score is mapped onto the platform's 0-100 scale; the live outputs show +10 for sub-score 1, +20 for sub-score 2 and +30 for sub-score 3 (e.g. "Temperature 38.5 °C (NEWS2 band 38.1-39.0, +10)"). This replaces the self-defined thresholds of the first iteration (e.g. fever ≥ 38.5 °C = +20).

Parameter	NEWS2 band	Sub-score	Engine points
Temperature (°C)	≤ 35.0 35.1-36.0 36.1-38.0 38.1-39.0 ≥ 39.1	3 1 0 1 2	30 10 0 10 20
Heart rate (bpm)	≤ 40 41-50 51-90 91-110 111-130 ≥ 131	3 1 0 1 2 3	30 10 0 10 20 30
SpO ₂ (%) — Scale 1	≤ 91 92-93 94-95 ≥ 96	3 2 1 0	30 20 10 0
Systolic BP (mmHg)	≤ 90 91-100 101-110 111-219 ≥ 220	3 2 1 0 3	30 20 10 0 30

Table 7.2 — NEWS2 reference bands and the observed $\times 10$ mapping to engine points.

Documented partial alignment

NEWS2 also scores respiratory rate and level of consciousness, which the intake form does not currently capture. This limitation is stated in the module's docstring rather than left implicit. qSOFA has not been implemented.

7.4 Risk levels

Level	Score threshold	Interpretation
LOW	< 15	Low risk
MODERATE	15 - 39	Moderate risk (stored as “medium”)
HIGH	40 - 69	High risk
CRITICAL	≥ 70	Critical risk

Threshold boundaries (0, 14, 15, 39, 40, 69, 70, 100) are covered by automated tests.

7.5 Model confidence

The first iteration displayed a fixed “Model Confidence: 60%” (a constant: 0.6 if factors were present, otherwise 0.35). It is now a **computed, five-factor score**. It is explicitly *not* a probability that the assessment is clinically correct: it reflects how much, and how reliable, the information behind the score is. It is capped at 95% so the tool never presents an assessment as a certainty.

Factor	Weight	What it measures
Vital-sign completeness	45%	How many of the 4 vitals were provided
Vital-sign validity	20%	Provided vitals are within a plausible physiological range
NLP entity yield	20%	0 / 1 / 2+ clinical entities detected
Concept diversity	10%	How many distinct clinical concepts (not just mentions)
Negation handling	5%	Slightly penalizes heavily negated admissions

Worked examples (real output from `compute_risk()`, reproduced end to end through the intake form; see Chapter 18):

Scenario	Level	Confidence
No data at all (empty admission)	LOW	5%
One vital (temperature), one symptom (headache), no negation	MODERATE (score 15)	51%
Three vitals (temperature, heart rate, blood pressure), one symptom, no negation	MODERATE (score 25)	74%
Full critical case: chest pain, shortness of breath, dizziness, hypertension, heart attack; all 4 vitals provided and valid	CRITICAL	95% (capped)

Confidence tracks data quality independently of the risk level: a clinician sees not just a level but, for instance, “moderate, with only 51% confidence”, a signal that more vitals or a fuller history would sharpen the assessment.

7.6 Calculation example (critical case)

Factor	Source	Points
Chest pain	NLP entity	+35
Shortness of breath	NLP entity	+30
Dizziness	NLP entity	+10
Hypertension	NLP entity (history)	+5
Heart attack	NLP entity (history)	+15
Heart rate 118 bpm	NEWS2 band 111-130	+20
SpO ₂ 90.5 %	NEWS2 band ≤ 91	+30
Cumulative	Result: CRITICAL · confidence 95% (capped)	145 → capped at 100

8 AI Assistant

The AI Assistant answers clinician questions in two strictly scoped modes. It was scoped along the lines later recommended in the review: RAG over a curated knowledge base rather than a freeform chatbot, a visible non-substitutive disclaimer, full logging, and role-based scope.

8.1 Two-mode architecture

The assistant routes deterministically between two modes depending on whether a `patient_id` is supplied by the calling frontend context: never a freeform, unscoped LLM call, and never a mixed call.

Mode	Trigger	Data injected	Where
Patient context	<code>patient_id</code> present	Structured data already persisted: <code>patient_intakes</code> , <code>extracted_entities</code> , <code>risk_assessments</code> . It explains the persisted score; it does not recompute it.	Embedded on the risk-result page
General knowledge (RAG)	no <code>patient_id</code>	Only top-k chunks retrieved by vector search from the clinic's reference documents; mandatory inline source citation (e.g. [Source 1: hypertension.md]). Never reads a patient record.	Dedicated AI Assistant page

Generation is performed by a local Qwen model served through Ollama.

8.2 Guardrails implemented

Requested guardrail	Implementation	Status
RAG over a curated knowledge base, not a freeform chatbot	Grounding, not memory: only retrieved chunks or persisted structured data are injected	DONE
Non-substitutive disclaimer on every response	Same disclaimer as the rest of the platform, visible under the input box	DONE
Every interaction logged (question, answer, sources)	<code>ai_conversations</code> table: question, answer, mode, sources, user, patient	DONE
Different access scopes for patients vs. doctors	Both modes are doctor/admin only, with the same clinic-isolation rule (<code>_can_access_patient</code>). A patient-facing general mode is deferred pending a separate risk review	DONE
Design note before implementation	Written retroactively, because the feature was built before the feedback arrived	PARTIAL

Why this matters

The patient-context answer is grounded entirely in structured data already computed by the risk engine, and the general answer is traceable to a named source file, so the clinician can verify it against the clinic's own documentation rather than trusting the model's memory.

8.3 Asynchronous handling

The assistant's local LLM inference is the first component slow enough to justify asynchronous handling. The trade-off and a concrete migration path (FastAPI BackgroundTasks plus polling) are documented in `docs/async_assistant_note.md`. It is not implemented yet, by design: there is a single concurrent user per clinic.

9 Testing and Validation

9.1 Suite overview

An automated pytest suite of **37 tests, 37 passing** (Python 3.11, pytest 9.1.1) covers score capping, NEWS2 boundaries, concept deduplication, negation exclusion, confidence computation and negation edge cases.

File	Coverage	Tests
test_risk_scoring.py	Score capping, level boundaries, NEWS2 vitals, concept dedup, negation exclusion, model confidence	23
test_negation.py	EN/FR negation, clause breaks, documented limitations, sentence splitting, field-boundary regression	14
Total		37

9.2 Case study — a real bug caught by testing

Found during manual QA, confirmed and fixed with regression tests. When `symptoms_text` ended without a period (e.g. “No fever and no cough”), `medical_history` merged into the same sentence during NLP preprocessing (“...no cough history of hypertension...”). The negation window for “no cough” then extended into the next field, so hypertension and heart attack were misclassified as negated and silently excluded from the risk score.

Fix: field concatenation now goes through `_build_nlp_text()`, which enforces an explicit sentence boundary between the two fields regardless of the punctuation typed.

Before the fix	After the fix
hypertension and heart attack missing from Contributing Factors; the score of 100/100 did not count them	Both correctly present: hypertension (+5), heart attack (+15)

Covered by three tests in `TestFieldConcatenationBoundary`:

`test_missing_period_no_longer_leaks_negation`, `test_empty_medical_history_handled` and `test_none_medical_history_handled`. Such an omission would have silently under-counted risk for any admission whose symptom text lacks a final period, an easy thing for a real user to do.

10 Database Architecture

MySQL with SQLAlchemy as the ORM. Nine main entities linked by foreign keys ensure referential integrity and multi-clinic isolation.

Table	Role and key fields
clinics	Healthcare establishment. id (UUID), name (unique).
users	Accounts (patient, doctor, admin). id, full_name, email (unique), password_hash, role, clinic_id, phone, is_active.
patients	Patient record. id (UUID), user_id (nullable for direct admission), created_by, clinic_id, identity, blood_group, height, weight, emergency contact.
patient_intakes	Clinical admission. id (UUID), patient_id, reason, symptoms, history, treatments, allergies, vital signs, status.
extracted_entities	NLP entities for an admission. intake_id, entity_type, entity_value, confidence, negated (kept for audit, excluded from score if negated).
risk_assessments	Scoring result. intake_id, risk_score, risk_level (enum: low/medium/high/critical), explanation, ai_confidence (computed), reviewed_by.
clinical_notes	Doctor observations. patient_id, doctor_id, note, created_at.
appointments	Scheduling. patient_id, doctor_id, appointment_date, status, notes (reason).
ai_conversations	New. Audit log of every AI Assistant exchange: question, answer, mode, sources, user, patient.

10.1 Relationships

Users are associated with patients and with the operations they perform (record creator, doctor of an admission or appointment, note author). Admissions belong to patients. Extracted entities and risk assessments belong to admissions. Clinical notes and appointments belong to patients, with attribution to the doctor. AI conversations reference the user and, in patient-context mode, the patient. Every sensitive record carries a clinic_id.

11 Security and Access Control

11.1 Authentication

After bcrypt password verification and an active-account check, the backend issues a signed JWT (HS256) containing the user identifier with a configurable validity, sent in the Authorization: Bearer header of each request.

11.2 Authorization

Authentication alone is insufficient: each sensitive endpoint also verifies the role (direct admission, doctor dashboard, administration and the AI Assistant require a doctor or administrator).

Defense in depth: the frontend is never the sole barrier. All permissions are verified on the backend, and a direct API request is systematically re-validated.

11.3 Data isolation

Every query applies a systematic filter on the current user's clinic (patients, admissions, appointments, notes, assistant), preventing cross-clinic access.

11.4 Compliance basics — current status (review point 5)

Because the platform handles health data, the following items are stated plainly rather than partially claimed:

Item	Current state	Next step	Status
HTTPS end-to-end	Not applicable to the localhost development environment	Enforce at deployment: Nginx reverse proxy with Let's Encrypt (or a managed platform with HTTPS by default)	OPEN
Encryption at rest	Not implemented for sensitive fields (symptoms_text, medical_history)	Evaluate column-level encryption (sqlalchemy-utils EncryptedType) as a first step	OPEN
Data retention policy	Not yet written	Requires a product decision first (retention window, applicable local regulation)	OPEN

12 Frontend Architecture

12.1 Code organization

Layer	Role
pages/	Screens: Login, Register, Dashboard, DoctorDashboard, AddPatient, Appointments, Notifications, Settings, Support, AdminPanel, RiskResult, PatientForm, AI Assistant.
components/	Reusable components: Layout, ProtectedRoute, RoleHomeRedirect, VitalLine.
contexts/	Global state: AuthContext (session), ToastContext (transient notifications).
services/	API access layer, one module per resource, with snake_case (API) → camelCase (frontend) conversion.
types/	TypeScript interfaces mirroring backend schemas.
utils/	Cross-cutting functions (e.g. home page resolution by role).

12.2 Routing and access protection

react-router-dom associates each route with authorized roles through ProtectedRoute: redirect to /login if unauthenticated, to the role's home page if access is denied. RoleHomeRedirect sends users to their dashboard at the root.

12.3 Session management

AuthContext restores the session at load: a stored token is validated against /auth/me; on failure the session is reset and the user redirected to login.

13 Frontend-Backend Communication

REST/JSON: React → Axios → FastAPI → Services → SQLAlchemy → MySQL.

Method	Route	Description
POST	/auth/register	Account creation
POST	/auth/login	Login, returns a JWT
GET	/auth/me	Current user profile
PATCH	/auth/me	Profile update
PATCH	/auth/me/password	Password change
POST	/auth/forgot-password	Reset request
POST	/auth/reset-password	Reset via token
POST	/patients	Patient profile creation
GET	/patients/me	Current patient profile
GET	/patients	Clinic patients
GET/PUT/DELETE	/patients/{id}	Consultation, modification, deletion
POST	/patients/intake	Admission (patient)
POST	/patients/intake/staff	Direct admission (doctor/admin)
GET	/patients/intake	Accessible admissions
GET	/patients/intake/{id}	Admission details
GET	/patients/intake/{id}/entities	Extracted NLP entities
GET	/risk/{intake_id}	Risk assessment (score, level, factors, confidence)

Method	Route	Description
GET	/dashboard/patients	Doctor view (identity + risk)
GET	/dashboard/stats	Clinic statistics
GET	/patients/{id}/notes	Clinical notes
POST	/patients/{id}/notes	Write a clinical note
GET	/appointments/doctors	Clinic doctors
GET	/appointments/today	Today's appointments
GET	/appointments/upcoming	Upcoming appointments
POST	/appointments	Appointment creation

The AI Assistant (general and patient-context modes) adds its own endpoints, both restricted to doctor/admin and clinic-isolated.

14 Technology Stack

Component	Technology
Frontend	React 18 + TypeScript
Routing	React Router
HTTP client	Axios
UI	Tailwind CSS
Backend	FastAPI + Python 3.11
ORM	SQLAlchemy
Validation	Pydantic
Database	MySQL
Authentication	JWT (HS256) + bcrypt
ASGI server	Uvicorn
NLP	spaCy + medspaCy
NER	TargetRule / centralized clinical vocabulary
Negation	ConText (medspaCy) + multilingual window
Language detection	langdetect
Clinical scale	NEWS2 (vital-sign scoring)
AI Assistant	Local Qwen via Ollama; RAG with vector search over curated reference documents
Testing	pytest (37 tests)
API	REST / JSON

15 Complete Usage Scenario (A to Z)

- 1** The user logs in (or creates an account attached to a clinic).
- 2** The system verifies identity and issues a JWT; role permissions apply on every request.
- 3** The patient completes the administrative profile (identity, emergency contact...).
- 4** The patient provides the reason for consultation and describes symptoms.
- 5** Medical history, current treatments and allergies are provided; vital signs are added if available.
- 6** The backend validates the information (physiological bounds, required fields) and stores it in `patient_intakes`.
- 7** The clinical text is assembled with an explicit field boundary and sent to the NLP module, with automatic language detection.
- 8** The NLP extracts entities (symptoms, history, medications, allergies); the negation module marks each as present or negated.
- 9** Entities are persisted in `extracted_entities`, negated ones included for audit.
- 10** Retained factors (non-negated, non-zero weight) go to the risk engine, which combines them with NEWS2-scored vital signs.
- 11** The engine computes the score (capped at 100), the risk level, the readable explanation and the five-factor model confidence (capped at 95%).
- 12** The result is displayed in the doctor dashboard with a color code; admissions are grouped by emergency contact to flag potential duplicates.
- 13** The doctor consults the complete record (admission, extracted entities, score, confidence).
- 14** The doctor may ask the AI Assistant to explain the score (patient-context mode) or a general medical question (RAG mode); every exchange is logged.
- 15** The doctor confirms or adjusts the AI score; the decision is recorded in the AI-versus-clinician audit log.
- 16** The doctor can schedule a follow-up appointment and add a clinical note; the patient is notified.
- 17** The healthcare professional performs the final clinical review; the medical decision remains, under all circumstances, their sole responsibility.
- 18** In parallel, the administrator has a consolidated view (statistics, account management, audit log).

16 Operation Example

16.1 Data entered

A patient reports chest pain and shortness of breath, with dizziness; history of hypertension and a previous heart attack; all four vital signs provided, including heart rate 118 bpm and SpO₂ 90.5 %.

16.2 NLP extraction and assessment

Entity / vital	Status	Points
Chest pain	Present	+35
Shortness of breath	Present	+30
Dizziness	Present	+10
Hypertension	Present	+5
Heart attack	Present	+15
Heart rate 118 bpm	NEWS2 band 111-130	+20
SpO ₂ 90.5 %	NEWS2 band ≤ 91	+30

Result

Risk score 100 / 100 (capped) · CRITICAL · Model confidence 95% (capped).

16.3 Interpretation

The contributing factors are displayed to the doctor for immediate understanding: “chest pain (+35); shortness of breath (+30); dizziness (+10); hypertension (+5); heart attack (+15); Heart rate 118 bpm (NEWS2 band 111-130, +20); Low oxygen saturation 90.5% (NEWS2 band ≤91, +30)”. The 95% confidence tells the doctor the assessment rests on complete, valid data. The doctor then confirms or adjusts the score, and remains responsible for the final clinical interpretation.

17 Response to Review Feedback — Point by Point

The review of the first progress report highlighted strengths (layered architecture, multilingual NLP with negation, explainable risk engine, AI-versus-clinician audit log, documentation quality) and six areas to address, plus guidance on the AI Assistant. Each is answered below.

17.1 Point 1 — Model confidence **DONE**

The fixed value (0.6 if factors else 0.35) has been replaced by the computed, weighted five-factor confidence of Section 7.5. Real outputs: 5%, 51%, 74% and 95% (capped), reproduced end to end in the running application.

17.2 Point 2 — Clinical grounding of risk weights **DONE**

Vital-sign scoring is derived from NEWS2 (Royal College of Physicians, 2017), with each parameter's 0-3 sub-score mapped onto the 0-100 scale. This is an explicit **partial** alignment: respiratory rate and consciousness are not captured by the intake form. Symptom and history weights remain internally defined, since NEWS2 does not cover patient-reported symptoms. qSOFA was not implemented.

17.3 Point 3 — French/Arabic NLP depth **PARTIAL**

The limitation was already documented in `nlp_service.py` and is now verifiable through two regression tests (negation beyond the window, double negation). The optional lightweight multilingual clinical NER exploration has **not been started**; it is a candidate for the next iteration.

17.4 Point 4 — Testing and validation **DONE**

A 37-test automated suite (`test_risk_scoring.py`, `test_negation.py`) covers score capping, NEWS2 boundaries, concept deduplication, negation exclusion and confidence. Writing it surfaced a real production bug (Section 9.2), now fixed and covered by regression tests.

17.5 Point 5 — Security/compliance basics **OPEN**

Not yet addressed, stated plainly: HTTPS (to be enforced at deployment via Nginx and Let's Encrypt), encryption at rest for `symptoms_text` and `medical_history` (column-level encryption under evaluation) and the data-retention policy (a product decision is required first). See Section 11.4.

17.6 Point 6 — Synchronous pipeline **PARTIAL**

The review framed this as “fine for now”. The admission pipeline stays synchronous by design (well under a second; the frontend needs the result immediately). The AI Assistant's local inference is the first component slow enough to justify async handling; the trade-off and a migration path are documented, but not implemented (single concurrent user per clinic).

17.7 AI Assistant — design guardrails **DONE**

Every recommended guardrail (RAG over a curated base, disclaimer on every response, full logging, separate access scopes) was already in place; only the design note was written retroactively (Chapter 8).

17.8 Consolidated status

Area	Current state	Status
Model confidence	Five-factor computed formula, capped at 95%, verified with real worked examples	DONE
NEWS2 vital scoring	Temperature, heart rate, SpO2, systolic BP aligned; respiratory rate and consciousness not captured (documented)	DONE
Field concatenation bug	Found via manual QA, fixed with explicit sentence boundary, 3 regression tests	DONE
Test suite	37 tests, 37 passing	DONE
AI Assistant — general mode (RAG)	Working: retrieval + Qwen generation + source citation	DONE
AI Assistant — patient-context mode	Working: embedded on the risk-result page, doctor/admin only	DONE
AI Assistant — audit logging	ai_conversations table captures every exchange	DONE
Clinical notes	Doctor/admin write, patient + clinic staff read; stable	DONE
French/Arabic NLP depth	Limitation documented and test-verified; multilingual clinical NER not started (optional)	PARTIAL
Async handling	Pipeline synchronous by design; Assistant migration path documented, not implemented	PARTIAL
HTTPS enforcement	N/A on localhost; deployment plan documented (Nginx + Let's Encrypt)	OPEN
Encryption at rest	Not implemented for sensitive fields; column-level encryption identified as next step	OPEN
Data retention policy	Not yet written; requires a product decision first	OPEN

18 Screenshots of Interfaces

All screenshots come from a local test environment (localhost:5173) with demo accounts and data.

18.1 Authentication

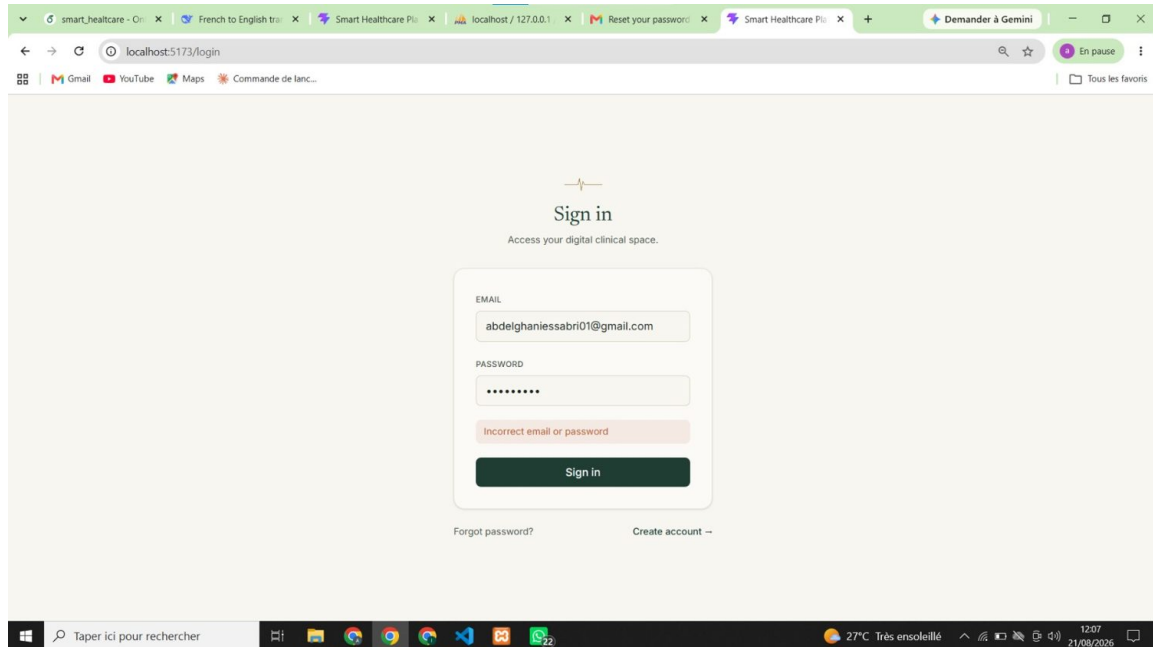


Figure 18.1 — Login (/login), with error message for incorrect credentials.

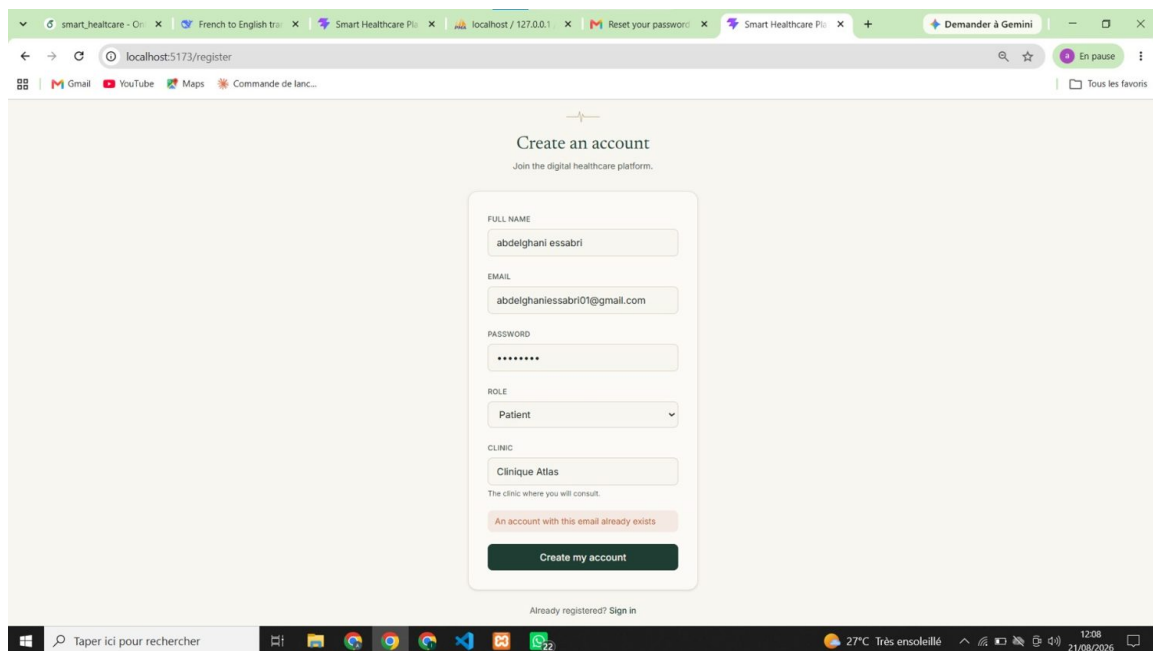


Figure 18.2 — Account creation (/register): role selection and clinic attachment.

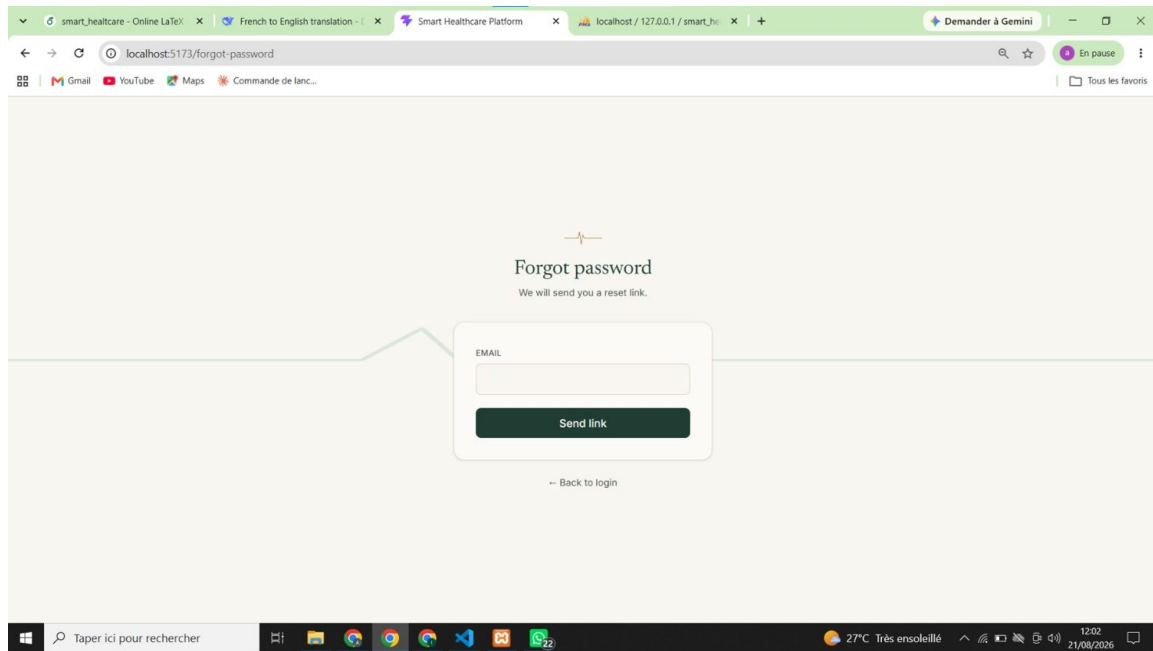


Figure 18.3 — Forgot password (/forgot-password).

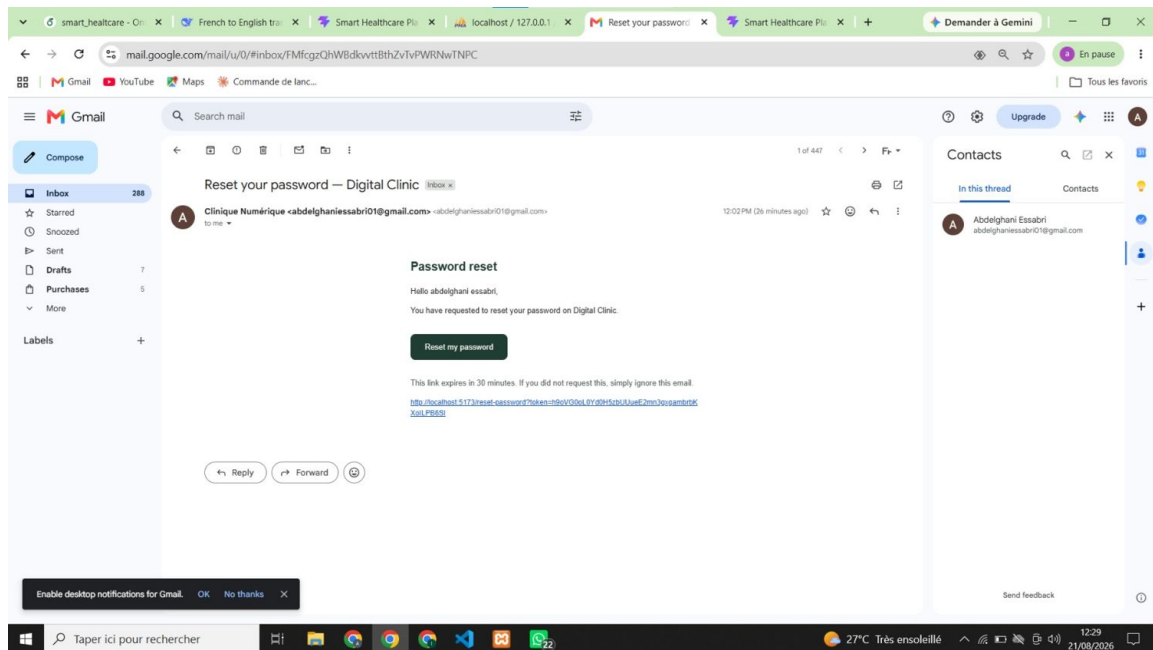


Figure 18.4 — Reset email received, link with limited lifetime (30 minutes).

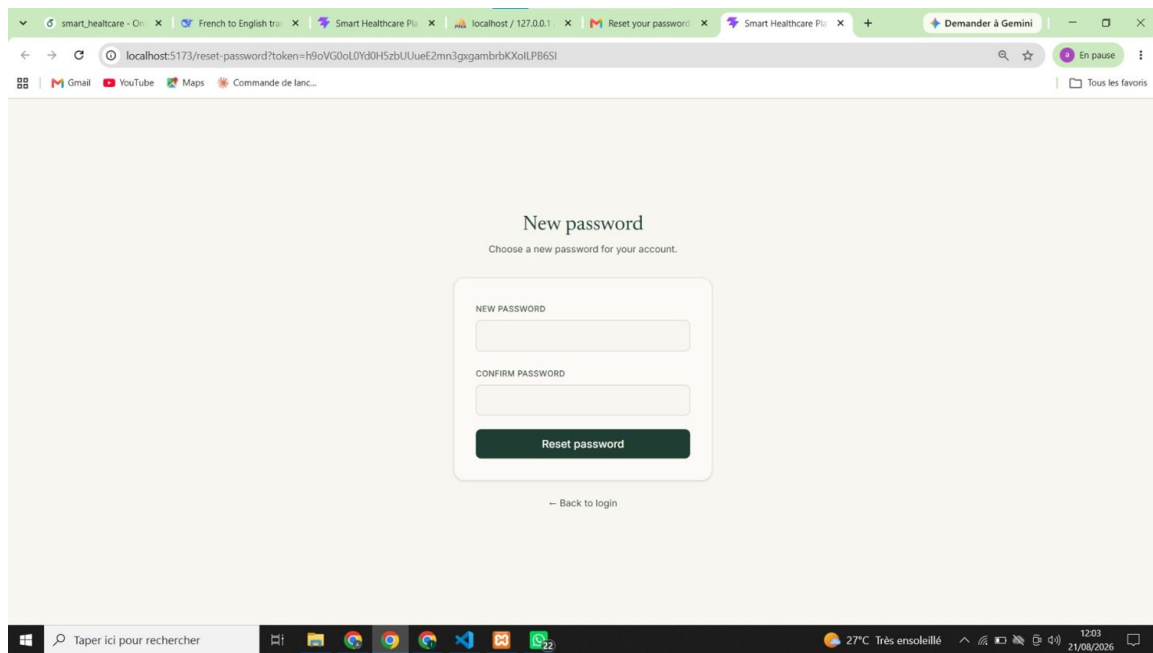


Figure 18.5 — Password reset (/reset-password?token=...).

18.2 Patient space

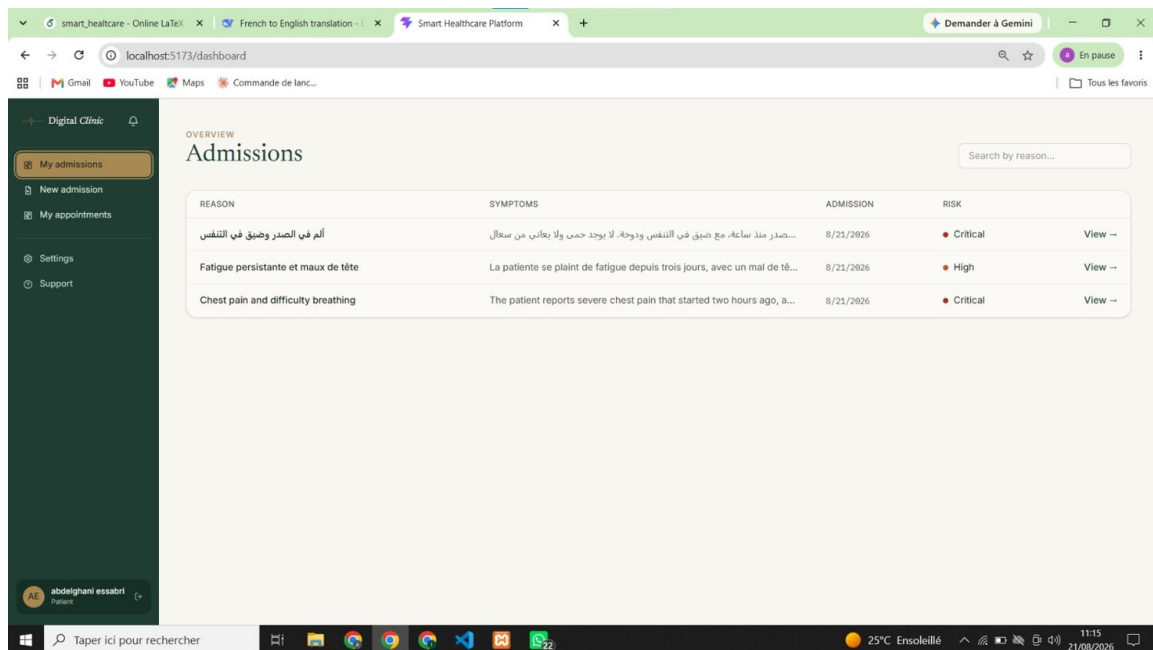


Figure 18.6 — Patient dashboard (/dashboard): admission history with risk level, including a case entered in Arabic.

The screenshot shows a web browser window with the URL `localhost:5173/patient-form`. The page is titled "CLINICAL RECORD" and "Patient admission". Below the title, it says "First complete your profile, then describe the reason for the visit." The form is divided into two main sections: "0 Patient profile" and "1 Reason for consultation".

0 Patient profile
This information is only requested once.

FIRST NAME: LAST NAME:

DATE OF BIRTH: GENDER:

BLOOD GROUP: HEIGHT (CM): WEIGHT (KG):

EMERGENCY CONTACT: EMERGENCY PHONE:

1 Reason for consultation
Describe the main reason for the consultation.

REASON FOR VISIT:

Figure 18.7 — New admission, step 0: patient profile.

The screenshot shows the same web browser window, but the form is now at step 2. The "Reason for visit" section is filled with "E.g. Abdominal pain". The "Current symptoms" section is empty. The "Medical history and treatments" section is also empty.

REASON FOR VISIT

CURRENT SYMPTOMS
Describe the symptoms, duration, progression and intensity...

2 Medical history and treatments
Relevant medical information for the assessment.

MEDICAL HISTORY
Chronic conditions, surgeries, hospitalizations...

CURRENT MEDICATIONS
Medication names and dosage if known...

ALLERGIES
Medication, food or other allergies...

Figure 18.8 — Steps 1 and 2: reason, symptoms and medical history.

smart_healthcare - Online LaTeX | French to English translation | Smart Healthcare Platform | Demander à Gemini

localhost:5173/patient-form

Medication names and dosage if known...

ALLERGIES

Medication, food or other allergies...

3 Vital signs

Measurements taken upon admission, if available.

TEMPERATURE: 37.0 °C

BLOOD PRESSURE: 120/80

HEART RATE: 72 bpm

O₂ SATURATION: 98 %

4 Additional information

Anything else useful for the clinician.

ADDITIONAL NOTES

Any additional information useful for the healthcare professional...

The information entered will be used to prepare the clinical assessment.

Submit record

abdelghani essabri Patient

Taper ici pour rechercher

25°C Ensoleillé 10:56 21/08/2026

Figure 18.9 — Steps 3 and 4: vital signs, additional information and submission.

smart_healthcare - Online LaTeX | French to English translation | Smart Healthcare Platform | localhost:127.0.0.1/smart_healthcare | Demander à Gemini

localhost:127.0.0.1/smart_healthcare

OVERVIEW

Notifications

Mark all as read

- Nouvelle note du médecin
Dr. imane essabri a ajouté une note à votre dossier.
just now
- Nouveau rendez-vous
Dr. imane essabri a planifié un rendez-vous le 23/08/2026 à 14:30.
11 min ago

View all notifications →

Search by reason...

SYMPTOMS	ADMISSION	RISK
The patient reports severe chest pain that started two hours ago, ac...	8/21/2026	Critical

View →

abdelghani essabri Patient

Taper ici pour rechercher

26°C Ensoleillé 11:47 21/08/2026

Figure 18.10 — Quick preview of unread notifications.

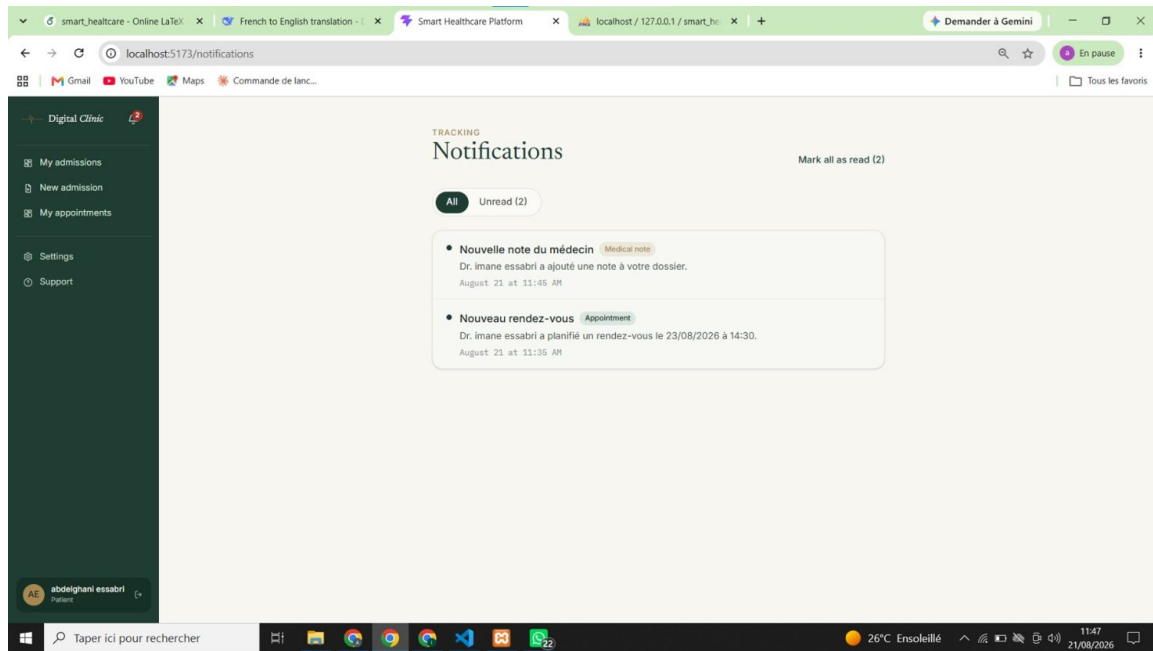


Figure 18.11 — Notification center (/notifications): all/unread filter and global marking.

18.3 Doctor space

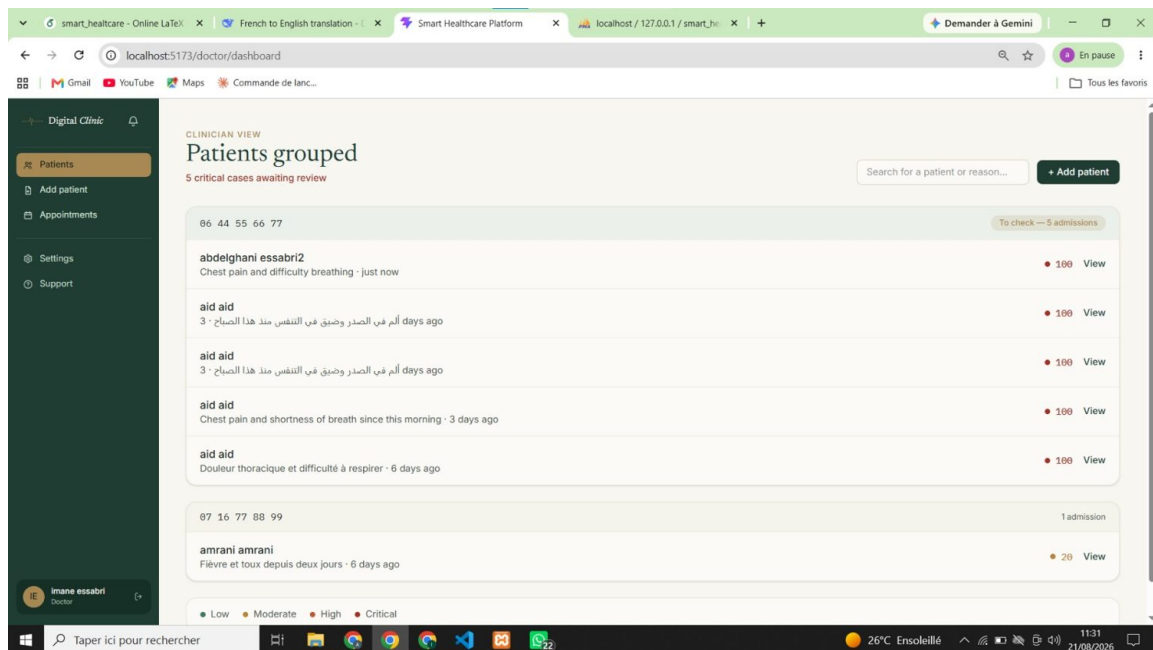


Figure 18.12 — Clinician view (/doctor/dashboard): patients grouped by contact, with risk color coding.

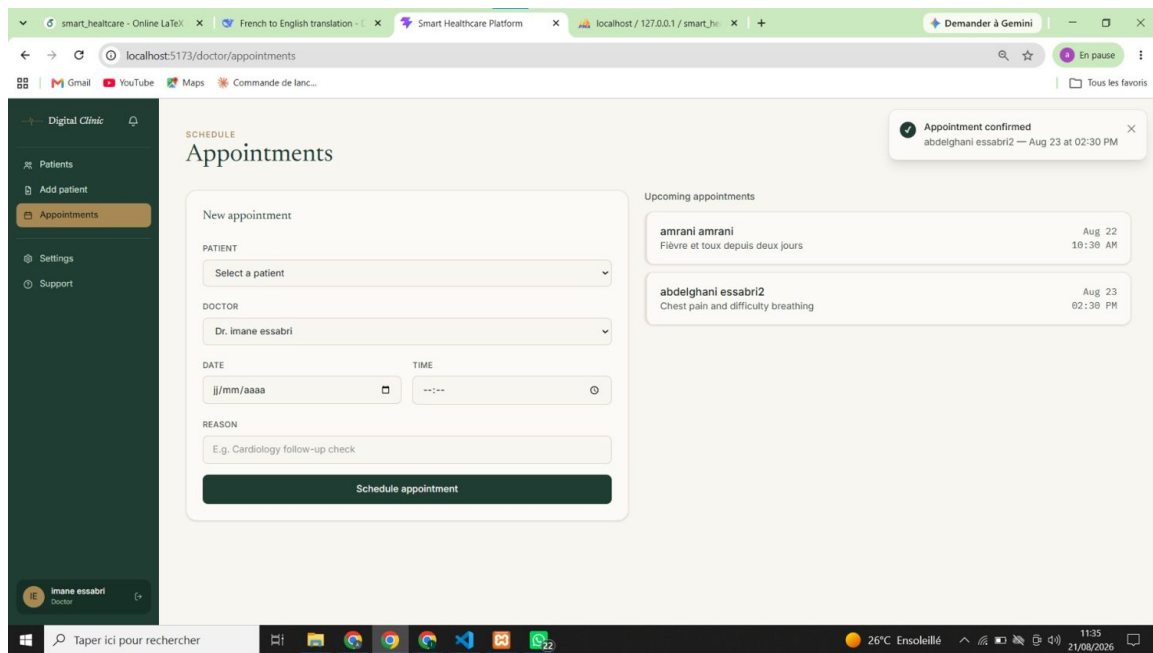


Figure 18.13 — Appointment management (/doctor/appointments).

18.4 Risk assessment and computed confidence (Iteration 2)

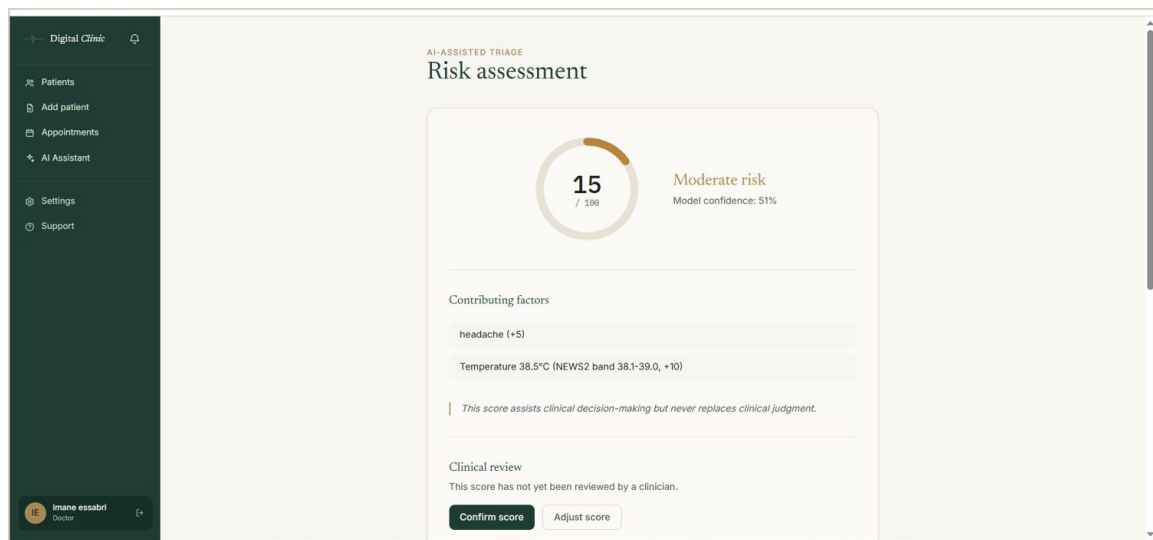


Figure 18.14 — One vital (temperature) and one symptom (headache), no negation: 51% confidence, as predicted analytically.

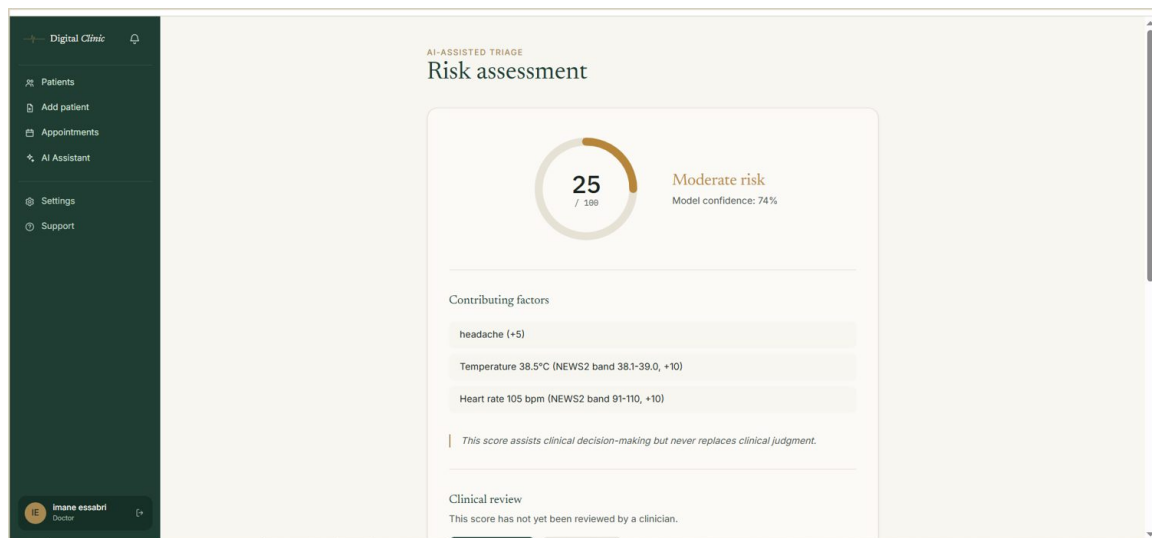


Figure 18.15 — Temperature, heart rate and blood pressure provided, one symptom, no negation: 74% confidence, as predicted. Blood pressure is within the normal range, hence it adds no contributing factor.

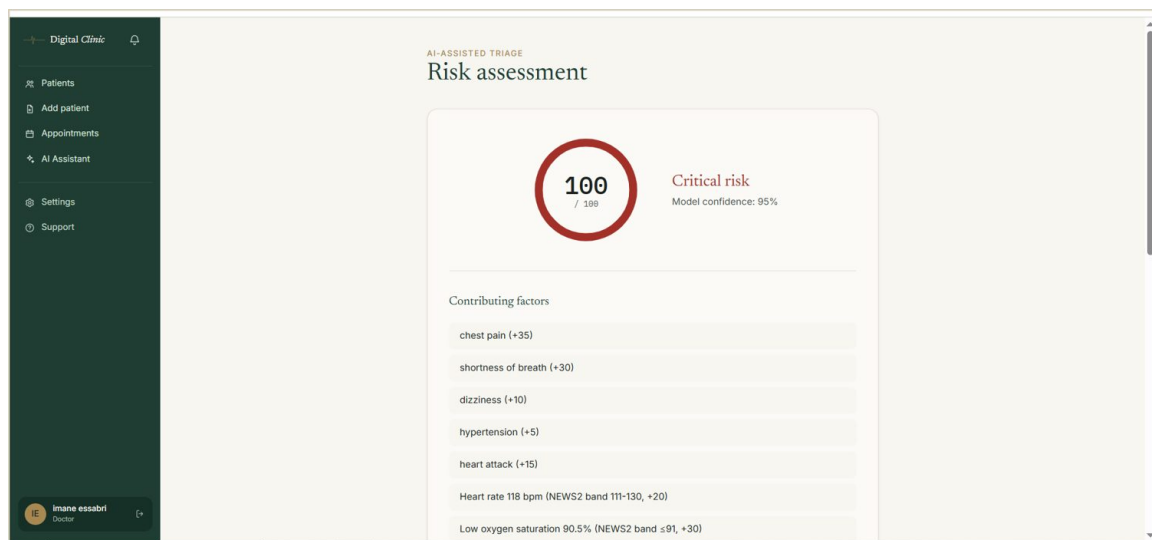


Figure 18.16 — Full critical case, all 4 vitals provided: 95% confidence (capped), with NEWS2-labelled vital factors.

18.5 AI Assistant

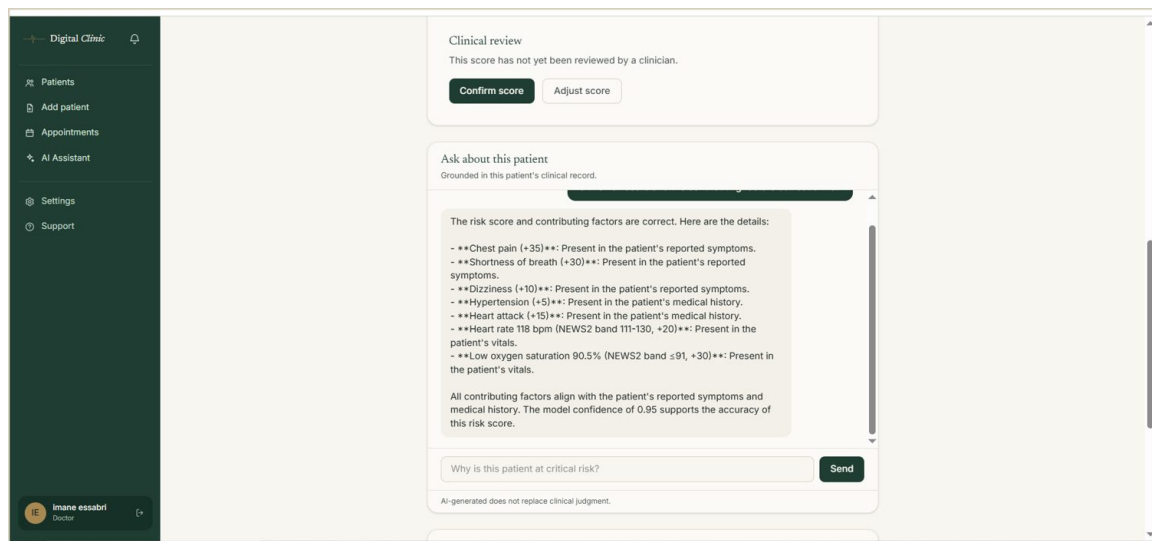


Figure 18.17 — Patient-context mode, embedded on the risk-result page: explains each contributing factor from the persisted assessment and cites the same 0.95 confidence.

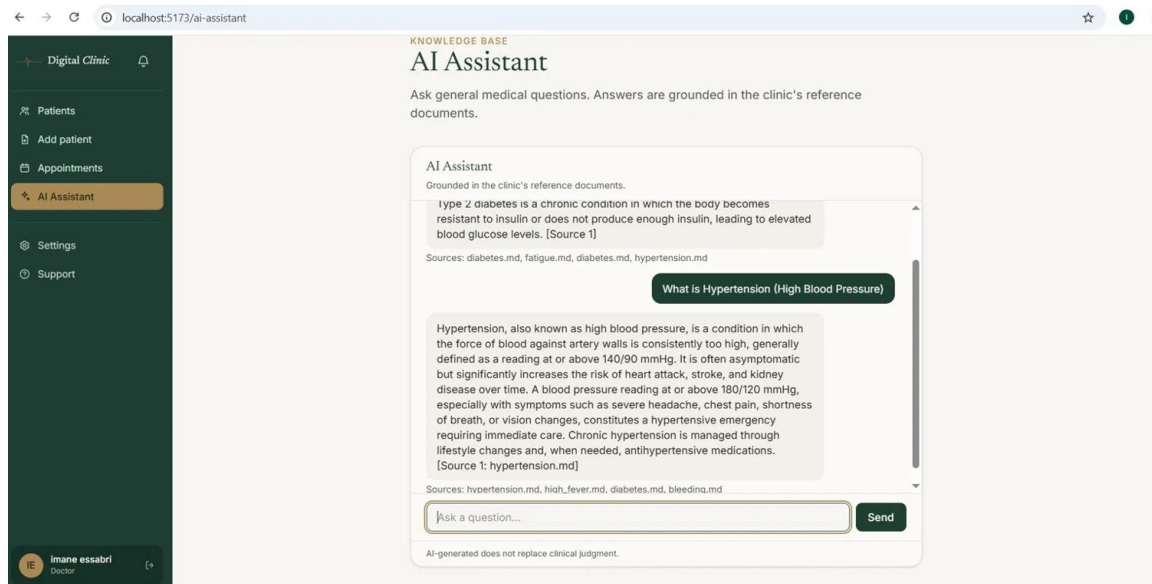


Figure 18.18 — General mode (RAG): answer grounded in the clinic's reference documents, inline citation, retrieved files listed, persistent disclaimer.

18.6 Administration

The screenshot shows the 'Administration' page in the Smart Healthcare Platform. The page title is 'Administration' with a subtitle 'Read-only overview of the clinic — patients, appointments, team and clinical decision history.' The page features four summary cards: 'USERS' (9), 'DOCTORS' (1), 'PATIENTS' (3), and 'APPOINTMENTS' (3). Below these is a table titled 'Users and roles' with columns: NAME, EMAIL, ROLE, and STATUS. The table lists 10 users with their roles (Admin, Patient, Doctor) and status (Active, Disabled). A sidebar on the left contains links for 'Patients', 'Administration', 'Settings', and 'Support'. The bottom of the page shows a Windows taskbar with various application icons and system information.

NAME	EMAIL	ROLE	STATUS
abdelghani essabri	abdelghani01@gmail.com	Admin	Active
abdelghani essabri	abdelghaniessabri01@gmail.com	Patient	Active
ahmed aid	ahmed01@gmail.com	Patient	Disabled
ahmed hassan	mohamed01@gmail.com	Admin	Active
aid aid	aid01@gmail.com	Patient	Active
amrani amrani	amrani@gmail.com	Patient	Active
essabri abdelghani	essabri01@gmail.com	Patient	Disabled
essabri abdelghani	abdelghani02@gmail.com	Admin	Active
imane essabri	imane01@gmail.com	Doctor	Active

Figure 18.19 — Administration (/admin): clinic statistics and user list with role and status.

The screenshot shows the 'Administration' page with a focus on appointments and the audit log. The 'All appointments' section lists three scheduled appointments with details on the doctor, patient, and time. The 'Audit log – AI versus clinician' section shows two entries where Dr. Imane Essabri confirmed or adjusted AI scores for patients Amrani Amrani and Aid Aid. The sidebar and taskbar are consistent with the previous screenshot.

Appointment	Doctor	Patient	Status	Time
abdelghani essabri2 Dr. Imane Essabri - Chest pain and difficulty breathing	Imane Essabri	Abdelghani Essabri2	Scheduled	Aug 23, 02:30 PM
amrani amrani Dr. Imane Essabri - Fièvre et toux depuis deux jours	Imane Essabri	Amrani Amrani	Scheduled	Aug 22, 10:30 AM
aid aid Dr. Imane Essabri - Douleur thoracique et difficulté à respirer	Imane Essabri	Aid Aid	Scheduled	Mar 17, 10:30 AM

Audit Log Entry	AI Score	Adjusted Score	Time
Dr. Imane Essabri confirmed the score of amrani amrani AI score: 20 (moderate) — no modification	20 (moderate)	-	Aug 14, 11:05 PM
Dr. Imane Essabri adjusted the score of aid aid AI score: 100 (critical) — adjusted: 80 (critical)	100 (critical)	80 (critical)	Aug 14, 10:44 PM

Figure 18.20 — All appointments and the AI-versus-clinician audit log.

18.7 Automated test suite

```
(venv311) PS D:\Clinical_AI\smart-healthcare-backend> python -m pytest tests/ -v
===== test session starts =====
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0 -- D:\Clinical_AI\smart-healthcare-backend\venv311\Scripts\python.exe
cachedir: .pytest_cache
rootdir: D:\Clinical_AI\smart-healthcare-backend
plugins: anyio-4.15.1
collected 37 items

tests/test_negation.py::TestNegationEnglish::test_simple_negation_detected PASSED [ 2%]
tests/test_negation.py::TestNegationEnglish::test_no_negation_when_absent PASSED [ 5%]
tests/test_negation.py::TestNegationEnglish::test_negation_stops_at_clause_break_but PASSED [ 8%]
tests/test_negation.py::TestNegationEnglish::test_negation_stops_at_comma PASSED [ 10%]
tests/test_negation.py::TestNegationFrench::test_pas_de_detected PASSED [ 13%]
tests/test_negation.py::TestNegationFrench::test_sans_detected PASSED [ 16%]
tests/test_negation.py::TestNegationFrench::test_negation_stops_at_mais PASSED [ 18%]
tests/test_negation.py::TestNegationKnownLimitations::test_LIMITATION_negation_beyond_window_is_missed PASSED [ 21%]
tests/test_negation.py::TestNegationKnownLimitations::test_LIMITATION_double_negation_not_handled PASSED [ 24%]
tests/test_negation.py::TestSentenceSplitting::test_splits_on_period PASSED [ 27%]
tests/test_negation.py::TestSentenceSplitting::test_splits_on_multiple_delimiters PASSED [ 29%]
tests/test_negation.py::TestFieldConcatenationBoundary::test_missing_period_no_longer_leaks_negation PASSED [ 32%]
tests/test_negation.py::TestFieldConcatenationBoundary::test_empty_medical_history_handled PASSED [ 35%]
tests/test_negation.py::TestFieldConcatenationBoundary::test_none_medical_history_handled PASSED [ 37%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_never_exceeds_100 PASSED [ 40%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_never_negative PASSED [ 43%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[0-low] PASSED [ 45%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[14-low] PASSED [ 48%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[15-medium] PASSED [ 51%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[39-medium] PASSED [ 54%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[40-high] PASSED [ 56%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[69-high] PASSED [ 59%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[70-critical] PASSED [ 62%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[100-critical] PASSED [ 64%]
tests/test_risk_scoring.py::TestDeduplication::test_same_concept_counted_once PASSED [ 67%]
tests/test_risk_scoring.py::TestNegationExclusion::test_negated_entity_excluded_from_score PASSED [ 70%]
tests/test_risk_scoring.py::TestNegationExclusion::test_mixed_negated_and_present PASSED [ 72%]
tests/test_risk_scoring.py::TestNegationExclusion::test_zero_weight_entity_excluded PASSED [ 75%]
tests/test_risk_scoring.py::TestNEWS2Vitals::test_normal_vitals_contribute_zero PASSED [ 78%]
tests/test_risk_scoring.py::TestNEWS2Vitals::test_critical_oxygen_saturation PASSED [ 81%]
tests/test_risk_scoring.py::TestNEWS2Vitals::test_low_systolic_bp_flagged PASSED [ 83%]
tests/test_risk_scoring.py::TestNEWS2Vitals::test_malformed_blood_pressure_ignored_gracefully PASSED [ 86%]
tests/test_risk_scoring.py::TestModelConfidence::test_confidence_baseline_with_no_data PASSED [ 89%]
tests/test_risk_scoring.py::TestModelConfidence::test_confidence_increases_with_data_completeness PASSED [ 91%]
tests/test_risk_scoring.py::TestModelConfidence::test_confidence_never_exceeds_095 PASSED [ 94%]
tests/test_risk_scoring.py::TestModelConfidence::test_invalid_vital_does_not_count_as_quality PASSED [ 97%]
tests/test_risk_scoring.py::TestModelConfidence::test_single_vs_multiple_entities_changes_nlp_score PASSED [100%]

===== 37 passed in 1.54s =====
(venv311) PS D:\Clinical_AI\smart-healthcare-backend>
```

Figure 18.21 — `python -m pytest tests/ -v`: 37 of 37 tests passing on the actual project.

19 Limitations and Next Steps

19.1 Known limitations

- NEWS2 alignment is partial: respiratory rate and level of consciousness are not captured; qSOFA is not implemented.
- Symptom and history weights are internally defined (NEWS2 does not cover patient-reported symptoms).
- French and Arabic NLP use keyword and negation-window matching, with a likely higher false-negative rate than English; negation beyond the window and double negation are not handled.
- Model confidence measures data completeness and quality, not clinical correctness.
- The AI Assistant is doctor/admin only; a patient-facing general mode is deferred pending a separate risk review. Its design note was written retroactively.
- HTTPS, encryption at rest and a retention policy are not in place.

19.2 Next steps

- Enforce HTTPS at deployment (Nginx reverse proxy with Let's Encrypt, or a managed platform with HTTPS by default).
- Introduce column-level encryption for symptoms_text and medical_history; write the data-retention policy once the retention window and applicable regulation are decided.
- Explore a lightweight multilingual clinical NER for French and Arabic.
- Migrate AI Assistant inference to asynchronous handling (BackgroundTasks plus polling) if concurrency grows.
- Capture respiratory rate and consciousness level to complete the NEWS2 alignment.
- Perform a dedicated risk review before any patient-facing assistant mode.

20 Conclusion

Smart Healthcare combines patient management, multilingual medical language processing, explainable and clinically grounded risk assessment, a scoped AI Assistant and clinician review into one traceable chain:

Admission → Validation → NLP → Clinical extraction → Risk engine (NEWS2 + confidence) → Dashboard → Clinician review (confirm / adjust) → Audit log

This iteration turned six review points into a shipped fix, a documented and justified trade-off, or an honestly flagged open item; no point was left ambiguous. The NEWS2-aligned scoring and the computed confidence strengthen the claim to be a genuine decision-support tool rather than a black box: every number the clinician sees is traceable to a cited scale or a transparent formula. The test suite paid for itself immediately by surfacing a real scoring bug.

Clinical Decision Support

The intelligence added to this platform explains and grounds the system's output; it does not replace clinical judgment. The clinical decision remains, in every case, the responsibility of the healthcare professional.

Prepared for the XelerondAI team — Smart Healthcare Platform, consolidated report (Iterations 1 & 2), 2026.